

Digital Logic Design

CSE - 260

Assignment 03

Summer 2025

Note that:

- 1. You will have to submit the hard copy, show all steps in all questions*
- 2. Make an cover page which contains you name, student ID and section*
- 3. You do not need to submit the ungraded portion.*

Deadline: 23 August, 2025

Graded

1. Batman is chasing Joker. Two 3-bit numbers A and B represent Batmobile speed and Joker's speed:
 - a. If the sum of their LSBs $> 0 \rightarrow$ Accelerate ($A + B$)
 - b. Otherwise $\rightarrow$ Brake hard ($B - A$)

Ungraded

1. Design a circuit using 4-bit parallel adders to add four 4-bit numbers.
2. Design a circuit that takes two 5-bit numbers and outputs:
 - a. Sum if both are even
 - b. Difference if one is odd and the other is even
 - c. Zero if both are odd.
3. Inside Avengers Tower, Tony Stark is running a dangerous Arc Reactor test. Jarvis has measured two energy frequencies: A from the stabilizing coils and B from the main power core.

Tony's rule is simple:

 - a. If both readings are even or odd numbers in the 4-bit range, the system boosts the core by adding them.
 - b. Otherwise, it risks instability, so the system reduces load by subtracting B from A.